MID-SEMESTER TEST
(SOLUTION)

Q1) 4, 3, 5, 2, 1

$n = 5$, $\sum x_i = 15$, $\sum x_i^2 = 55$

$$s^2 = \frac{1}{(n-1)}\left\{\sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}\right\}$$

$$\therefore s^2 = \frac{1}{(5-1)}\left\{55 - \frac{(15)^2}{5}\right\}$$

$$= \frac{1}{4}\{55 - 45\} = \frac{10}{4} = 2.5$$

$$\therefore s = \sqrt{2.5} = \underline{\underline{1.58}}$$

Q2)

$$Q_1 = x_{[1+(n-1)0.25]} = x_{[1+(30-1)0.25]}$$

$$= x_{[8.25]} = (0.75)x_{(8)} + (0.25)x_{(9)}$$

$$= (0.75)(12.2) + (0.25)(13.6)$$

$$= 9.15 + 3.40$$

$$= 12.55$$

$$Q_3 = x_{[1+(n-1)0.75]} = x_{[1+(30-1)(0.75)]}$$

$$= x_{(22.75)}$$

$$= (0.25)x_{(22)} + (0.75)x_{(23)}$$

$$= (0.25)(23.2) + (0.75)(23.6)$$

$$= 5.8 + 17.7 =$$

$\therefore IQR = Q_3 - Q_1 = 23.5 - 12.55 = 10.95$

$1.5 \times IQR = 1.5 \times (10.95) = 16.425$

Hence, Lower fence

$Q_1 - (1.5 \times IQR) = 12.55 - 16.425 = -3.875$

Upper fence.

$Q_3 + (1.5 \times IQR) = 23.5 + 16.425 = 39.925$

Therefore outliers are only 40.9 and 42.1

No low outliers.

Question 3

$$f_X(x) = \begin{cases} \frac{3x^2}{\theta^3}, & 0 \leq x \leq \theta \\ 0, & \text{else} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^{\theta} x\left(\frac{3x^2}{\theta^3}\right)dx = \frac{3}{\theta^3}\int_0^{\theta} x^3\,dx$$

$$= \frac{3}{\theta^3}\left[\frac{x^4}{4}\right]_0^{\theta} = \frac{3}{\theta^3}\left[\frac{\theta^4}{4}\right] = \frac{3}{4}\theta$$

$\hat{\theta} = \bar{X}$, $E(\hat{\theta}) = E(\bar{X}) = E(X) = \frac{3}{4}\theta$

$BIAS = E\left[(\hat{\theta} - \theta)\right] = E(\hat{\theta}) - \theta = \frac{3}{4}\theta - \theta = -\frac{1}{4}\theta$

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Question 4

$$Y = c\bar{X}$$

$$MSE(Y) = [BIAS(Y)]^2 + VAR[Y]$$

$$X_i \sim iid\ Gamma\left(2, \frac{1}{\theta}\right)$$

$$E[Y] = E[c\bar{X}] = c\,E[\bar{X}] = c\,E[X] = c(2\theta)$$

$$\therefore E[Y] = 2c\theta, \quad BIAS(Y) = E[Y] - \theta$$

$$= 2c\theta - \theta$$

$$\therefore BIAS = (2c-1)\theta$$

$$VAR(Y) = VAR(c\bar{X}) = c^2\,VAR(\bar{X})$$

$$= c^2 \frac{VAR(X)}{n}$$

$$= \frac{c^2(2\theta^2)}{n} = \frac{2c^2\theta^2}{n}$$

$$MSE(Y) = [(2c-1)\theta]^2 + \frac{2c^2\theta^2}{n}$$

$$= (2c-1)^2\theta^2 + \frac{2c^2\theta^2}{n}$$

Question 5

$n = 8$, 6 censored observations

Non-censored.

$x_1 = 1.4, \; x_2 = 1.6$

$T \sim Exp(\lambda)$.

$\sum x_i = 1.4 + 1.6 = 3.0$

For uncensored case.

$$\prod_{i=1}^{2} \lambda e^{-\lambda x_i}$$

$$P(T > 2) = \int_2^{\infty} \lambda e^{-\lambda x} dx = \frac{\lambda e^{-\lambda x}}{(-\lambda)} \Big|_2^{\infty}$$

$$= \left[-e^{-\lambda x} \right]_2^{\infty} = \left[\frac{-1}{e^{\lambda x}} \right]_2^{\infty} = \frac{1}{e^{2\lambda}}$$

$$= e^{-2\lambda}$$

Hence Likelihood,

$$L = \left(\prod_{i=1}^{2} \lambda e^{-\lambda x_i} \right) \times \left(P(x_3, \ldots, x_8) > 2 \right)$$

$$= \left(\lambda^2 e^{-\lambda \sum_{i=1}^{2} x_i} \right) \times \left\{ \left[P(x_3) > 2 \right] \left[P(x_4 > 2) \right] \cdots \left[P(x_8 > 2) \right] \right\}$$

$$= \left(\lambda^2 e^{-\lambda \Sigma x_i} \right) \times \left(e^{-2\lambda} \right)^6$$

$$= \lambda^2 e^{-3\lambda} e^{-12\lambda}$$

$$= \underline{\lambda^2 e^{-15\lambda}}$$

Question 6

$X_i \sim \text{Beta}(\theta, 4)$.

From formula sheet if $X \sim \text{Beta}(\alpha_1, \alpha_2)$

then $E(X) = \dfrac{\alpha_1}{\alpha_1 + \alpha_2}$

Hence for

$X \sim \text{Beta}(\theta, 4)$, $E(X) = \dfrac{\theta}{\theta + 4}$

Method of Moments

Pop moments = Sample moment.

$$\frac{\hat{\theta}}{\hat{\theta} + 4} = \bar{X}$$

$$\hat{\theta} = \bar{X}(\hat{\theta} + 4)$$

$$= \bar{X}\hat{\theta} + 4\bar{X}$$

$$\hat{\theta} - \bar{X}\hat{\theta} = 4\bar{X}$$

$$\hat{\theta}(1 - \bar{X}) = 4\bar{X}$$

Hence $\hat{\theta}_{MME} = \dfrac{4\bar{X}}{1 - \bar{X}}$,

where $\bar{X}$ is sample mean.